Introduction to Web Technologies & HTML 5 Language

New Semantic& structural Elements In HTML 5

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HTML 5 Semantic & Structural Elements

❑ Semantic Means:

- When people say they want to make something more semantic, they simply want to make that thing more meaningful.
- Semantic html is using html to reinforce structural meaning. It's about using tags, class names, and ids that reinforce the meaning of the content within the tags.

❑ Why Semantic HTML is Important

- **Clean** — It's easier to read and edit, which saves time and money during maintenance.
- **More accessible** — It can be better understood by a greater variety of devices.
- **Search engine friendly** — This is still debatable as search engines rank content and not code.

HTML 5 Semantic & Structural Elements (Cont.)

- **Some New semantic Elements in HTML 5**
  - **Meter**
  - **Progress**
  - **Mark**
  - **Time**
  - **<header>**
  - **<hgroup>**
  - **<nav>**
  - **<section>**
  - **<article>**
  - **<aside>**
  - **<footer>**

<meter>

- **Representing scalar measurements or fractional values**
- **Meter is also known as a gauge**
- **It should not be used to indicate progress**
- **Attributes:**
  - value
  - min
  - max
  - high
  - low
  - optimum

<meter> (Cont.)

- **Using <meter>**

```
<meter value="0.6“
                    min=“”
                    max=“”
                    high ="0.6“ >
                    Medium
</meter>
```

<progress>

- **Show completion progress of a task**
- **Progress bars are widely used in other applications**
- **Works with scripted applications**
- **Attributes**
  - **value :** Specifies how much of the task has been completed
  - **max :** Specifies how much work the task requires in total
- **Useful for:**
  - Indicate loading progress of an AJAX application
  - Show user progress through a series of forms
  - Making impatient users wait

<progress> (Cont.)

- Using <progress>

```
<progress id=“pBar" max="100" value=“10">
</progress>
```

<mark>

- **Marked or Highlighted text**
- **Indicates point of interest or relevance**
- **Useful for:**
  - Highlighting relevant code in a code sample
  - Highlighting search keywords in a document (e.g. in Google Cache)

<mark> (Cont.)

- Using <mark>

The highlighted part below is where the error lies:

```
var i: Integer;
begin
   i := 1.1;
end.
```

```
<p>The highlighted part below is where the error lies:</p>
<code>
var<var> i</var>: Integer;
begin
i := <mark>1.1</mark>;
end.
</code>
```

is used for computer code

is used to indicate a variable within code.

<Time>

- Time as you would expect is used to markup temporal information. It can be used for dates, times, or combinations of the two.
- The <time> tag defines either a time (24 hour clock), or a date in the Gregorian calendar, optionally with a time and a time-zone offset.
- This element can be used as a way to encode dates and times in a machine-readable way.

```
<time>2011-07-14</time>
<time datetime="2011-07-14">July 14th, 2011</time>
<time datetime="2011-07-14T14:00">2pm on July 14th</time>
```

Structural Elements (Cont.)

Structural Semantic Elements:

- <header>
- <hgroup>
- <nav>
- <section>
- <article>
- <aside>
- <footer>
- <address>

More about Semantic & Structural Elements:

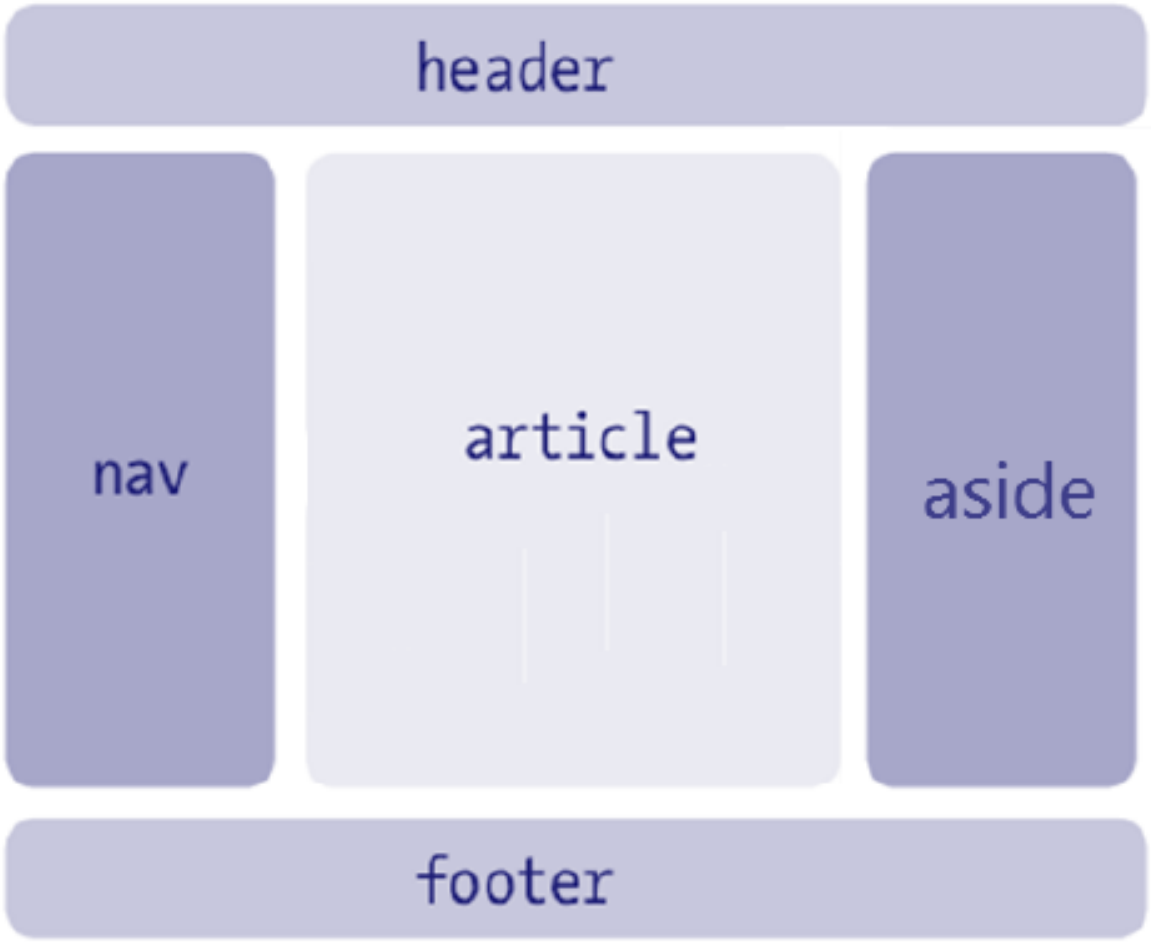
- http://www.adobe.com/devnet/dreamweaver/articles/understanding-html5-semantics.html
- http://www.webmonkey.com/2010/02/building_web_pages_with_html_5/

Structural Elements (Cont.)

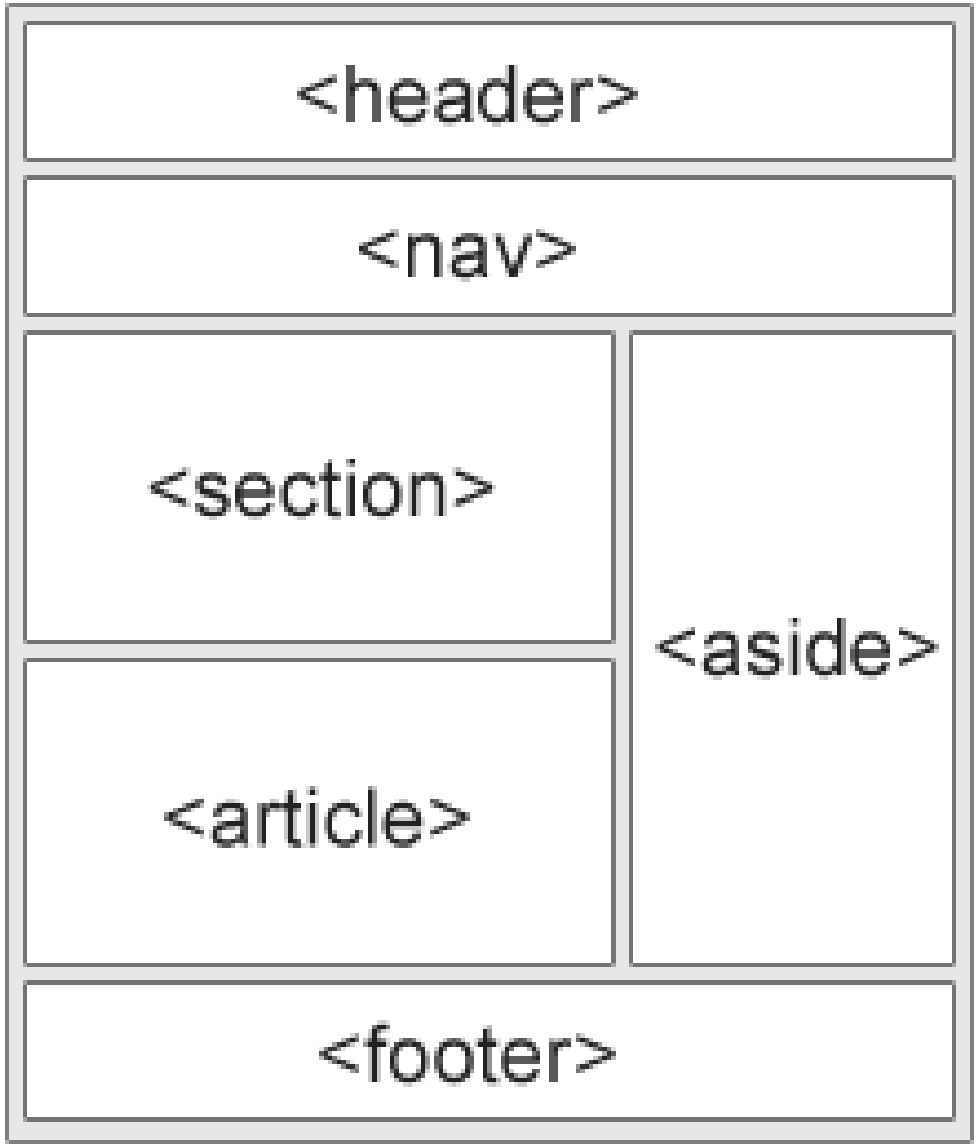
❑ Structural Semantic Usage:

Typical HTML4	Typical HTML5
	<header>
	<nav>
	<section>
	<article>
	<footer>

Structural Elements (Cont.)



Structural Elements (Cont.)



<Questions> ? </Questions>

Thank You…